

CSES — Counting Problems Reference

All tasks: 1 s, 512 MiB. Unless stated otherwise, answers are modulo $10^9 + 7$.

1075 — Counting Permutations

Count permutations p of $1, \dots, n$ such that no adjacent pair differs by 1, i.e. $|p_i - p_{i+1}| \neq 1$ for every i . **In:** $\mathbf{n}$

Out: number of beautiful permutations mod $10^9 + 7$

Constraints: $1 \leq n \leq 1000$.

1078 — Grid Paths II

In an $n \times n$ grid, move from $(1, 1)$ to (n, n) using only right/down moves while avoiding m trapped cells. Start/end are free and traps are distinct. **In:** $\mathbf{n \ m}$; then m traps $\mathbf{y \ x}$

Out: number of valid paths mod $10^9 + 7$

Constraints: $n \leq 10^6$, $m \leq 1000$, $1 \leq y, x \leq n$.

1080 — Empty String

Given a lowercase string, repeatedly remove any adjacent equal pair. Count the possible sequences of removals that delete the entire string. **In:** lowercase string s

Out: number of ways mod $10^9 + 7$

Constraints: $1 \leq |s| \leq 500$.

2176 — Counting Bishops

Place exactly k bishops on an $n \times n$ chessboard so that no two share a diagonal. Count all placements. **In:** $\mathbf{n \ k}$

Out: number of valid placements mod $10^9 + 7$

Constraints: $1 \leq n \leq 500$, $1 \leq k \leq n^2$.

2228 — Counting Sequences

Count length- n sequences over $\{1, \dots, k\}$ in which every value $1, \dots, k$ appears at least once. **In:** $\mathbf{n \ k}$

Out: number of sequences mod $10^9 + 7$

Constraints: $1 \leq k \leq n \leq 10^6$.

2229 — Permutation Inversions

Count permutations of $1, \dots, n$ having exactly k inversions, where an inversion is a pair $i < j$ with $p_i > p_j$. **In:** $\mathbf{n \ k}$

Out: number of permutations mod $10^9 + 7$

Constraints: $1 \leq n \leq 500$, $0 \leq k \leq n(n-1)/2$.

2415 — Functional Graph Distribution

A functional graph has labeled nodes and outdegree exactly 1 at every node. For every $k = 1, \dots, n$, count such graphs with exactly k connected components. **In:** $\mathbf{n}$

Out: n lines: answer for each $k = 1, \dots, n$ mod $10^9 + 7$

Constraints: $1 \leq n \leq 5000$.

2421 — Counting Reorders

Reorder the multiset of characters of a lowercase string so that adjacent characters are never equal. Count distinct resulting strings. **In:** lowercase string s

Out: number of valid reorderings mod $10^9 + 7$

Constraints: $1 \leq |s| \leq 5000$.

2429 — Grid Completion

Complete an $n \times n$ grid of $\cdot$, A, B so every row and column contains exactly one A and exactly one B. Existing letters are fixed; initially each row/column has at most one of each. **In:** $\mathbf{n}$; then n grid rows

Out: number of completions mod $10^9 + 7$

Constraints: $2 \leq n \leq 500$.

3157 — Collecting Numbers Distribution

For a permutation of $1, \dots, n$, repeatedly scan left-to-right, collecting the next required values in increasing order until the scan ends. For every k , count permutations requiring exactly k scans. **In:** $\mathbf{n}$

Out: n answers for $k = 1, \dots, n$, mod $10^9 + 7$

Constraints: $1 \leq n \leq 5000$.

3232 — Tournament Graph Distribution

A tournament has exactly one directed edge between each pair of labeled nodes. For every $k = 1, \dots, n$, count tournaments with exactly k strongly connected components. **In:** $\mathbf{n}$

Out: n lines: answer for each $k = 1, \dots, n$ mod $10^9 + 7$

Constraints: $1 \leq n \leq 500$.

3400 — Raab Game II

Each player owns cards $1, \dots, n$ and plays every card exactly once over n turns. Higher card scores 1 point, ties score neither; count ordered pairs of play orders giving final scores (a, b) . **In:** $\mathbf{t}$; then t tests $\mathbf{n \ a \ b}$

Out: answer for each test mod $10^9 + 7$

Constraints: $t \leq 1000$, $n \leq 5000$, $0 \leq a, b \leq n$.

3413 — Filled Subgrid Count I

Grid entries are the first k uppercase letters. For each letter, count contiguous axis-aligned *square* subgrids whose every cell equals that letter. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3414 — Filled Subgrid Count II

For each of the first k uppercase letters, count contiguous axis-aligned *rectangular* subgrids whose every cell equals that letter. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3415 — All Letter Subgrid Count I

Count contiguous axis-aligned *square* subgrids containing every one of the first k uppercase letters at least once. **In:** $\mathbf{n \ k}$; then n grid rows

Out: exact number of qualifying squares

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3416 — All Letter Subgrid Count II

Count contiguous axis-aligned *rectangular* subgrids containing every one of the first k uppercase letters at least once. **In:** $\mathbf{n \ k}$; then n grid rows

Out: exact number of qualifying rectangles

Constraints: $n \leq 500$, $k \leq 26$. Use 64-bit.

3417 — Border Subgrid Count I

For each letter, count contiguous axis-aligned *square* subgrids whose entire border consists of that letter; interior cells may be arbitrary. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3418 — Border Subgrid Count II

For each letter, count contiguous axis-aligned *rectangular* subgrids whose entire border consists of that letter; interior cells may be arbitrary. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 500$, $k \leq 26$. Use 64-bit.